

AISYS RFID Library Management System

Technical Evaluation & 7-Layer Modular Architecture

Presenter: **Mouli Palluru**

Repository: github.com/MouliPalluru/aisys_rfid_library

Target: support@aisys.co.in

49 / 49

Acceptance Tests Passed
(100% Pass Rate)

Executive Assessment & Core Metrics

Demonstrated

49 / 49

Pytest scenarios pass (~2.20 s)

20,000

records imported in < 5 s

- 49 / 49 Pytest scenarios pass (~2.20 s execution time).
- 20,000 records imported in under 5 seconds (CSV & XLSX supported).
- Modular FastAPI backend covering circulation, NCIP 2.0 / SIP2 adapters, RFID tag association, gate alarms, handheld audit and RBAC.

SOP Position

7 / 31

fully met

22

partially met

2

queued for backlog

Deliverables Index

- D1 Traceability matrix
- D2 / D3 Architecture specification
- D4 SBOM and licence inventory
- D5 Migration tool
- D6 Acceptance tests
- D7 / D8 Operations runbook
- D9 Presentation deck

7-Layer Modular Architecture

Modular Architecture Breakdown

- **Layer 1 • FastAPI / RBAC checks** — REST endpoints, validated requests, role checks
- **Layer 2 • NCIP 2.0 / SIP2** — mock protocol adapters over one shared rules engine
- **Layer 3 • Circulation Rules** — 5-item cap, ₹500 fine limit, member blocks, reference protection
- **Layer 4 • RFID Middleware & Gate Security** — tag association, security-bit check, alarms, mock CCTV
- **Layer 5 • Handheld Audit** — shelf scans flag missing, misplaced and unknown items
- **Layer 6 • Notifications** — queued security e-mails with background delivery
- **Layer 7 • Persistence Repository** — SQLite schema, staging table and backup for migrated data

Layer Stack

1 API & RBAC

2 Protocol Interoperability

3 Circulation Logic

4 RFID Middleware & Gate Security

5 Inventory Audit

6 Notifications

7 Persistence Data Layer

High-Throughput Data Migration (D5)

Ingestion & Reconciliation Engine (migrate.py)

Bulk Ingestion

- CSV and XLSX input through Pandas
- Members and book copies with RFID tags
- 20,000 rows imported in about 3 seconds

Schema Validation

- Required columns, email, phone, ISBN, tag UID, year
- Duplicates checked in file, staging table and database
- Rejected rows written to error_log.csv

Transactional Rollback

- One transaction per batch
- --rollback reverts the whole batch if any row is rejected
- Unexpected errors always revert; existing records are never modified

Reconciliation Logging

- Total Read, Valid, Rejected, Inserted
- Automatic consistency check on the counts
- Every row's outcome kept in the staging table

≈ 3 s

20,000 rows imported

0

existing records modified (insert-only)

1

transaction per batch

Circulation Logic & Security Gate Middleware

Circulation Rules

5 items

loan cap per member

₹500

fine limit at check-out

Blocks

member account blocking

Reference

item protection

Exit Gate Security

Gate event



Security-bit check



Alarm + accession no.



Mock CCTV capture



Queued e-mail

- Real-time RFID tag monitoring at the exit gate
- Unauthorized item detection: unissued or unknown tags fail secure
- Mock CCTV snapshot reference attached to every alarm

Protocol Adapters & Handheld Inventory Audit

Handheld Shelf Inventory

- Batch reconciliation of scanned tags against shelf records
- Flags missing and misplaced books
- Also reports items on loan but on the shelf, and unknown tags
- Duplicate reads are collapsed; the audit never alters records

Mocked handheld reader

Protocol Interoperability

- NCIP 2.0 adapter for inter-library workflows: CheckOutItem, CheckInItem, LookupItem
- SIP2 boundary for kiosk hardware: messages 11, 09 and 17
- Both adapters call the same circulation rules as the REST API

Simplified NCIP (JSON) and SIP2 mocks

Interactive API Documentation & Verification

Live API Control (Swagger UI)

Interactive Swagger UI

```
http://127.0.0.1:8000/docs
```

Try every endpoint in the browser while the service runs locally.

Role-Based Endpoint Execution

```
admin · librarian · patron
```

Send the X-Role header to exercise protected endpoints; other callers receive 403.

Strict JSON Schemas

```
Pydantic request models
```

Malformed requests are rejected with clear validation errors.

Demo tip: start the service with `uvicorn main:app --reload`, open `/docs`, and add the header `X-Role: admin` to try member blocking and the alarm log.

Access rules: admin: block members, view alarms; admin or librarian: associate tags; all other callers: 403.

Automated Verification & Quality Assurance (D6)

Automated Acceptance Suite (test_acceptance.py)

49

Passed

0

Failed

100%

Pass Rate

~2.20 s

Execution Time

Test Scenario Coverage

- Spreadsheet import, validation, duplicates and reconciliation (AC 01)
- RFID tag association and uniqueness (AC 02)
- NCIP 2.0 and SIP2 check-out / check-in flows (AC 03)
- Circulation blocking: reference, blocked member, fine limit (AC 04)
- Handheld inventory audit: missing, misplaced, unknown (AC 05)
- Gate event, accession number, mock CCTV, queued e-mail (AC 06)
- Role-based access control, 12 tests (AC 07)
- Migration rollback and failure recovery (AC 10)

Demo, Repository & Project Links

Everything is Reproducible

1 • Repository

```
github.com/MouliPalluru/aisys_rfid_library
```

Source code, tests, runbook and project documents

2 • Interactive API Docs

```
http://127.0.0.1:8000/docs
```

Available while uvicorn main:app is running

3 • Acceptance Test Command

```
pytest -v test_acceptance.py
```

49 tests in 2.20 s

4 • Migration Command

```
python migrate.py  
sample/members_sample.csv --entity members
```

Create demo data first: `python migrate.py --make-sample sample`

Production Readiness & Architectural Backlog

Production Roadmap & Next Engineering Steps

1

PostgreSQL Database Upgrade

Replace the in-memory API store and the SQLite migration target with one persistent PostgreSQL layer.

2

OAuth2 / JWT Authentication Hardening

Replace the header-based role check with verified identity, TLS and audit logging.

3

Web Admin Dashboard

Statistics and filterable reports for items, members, circulation, operators and gate events.

4

UAT Sign-off

Library-staff acceptance testing and Windows 11 / Server 2022 validation of the install, update and rollback procedures.